



Drupal 7: Features and Tracking Changes

The features module is great for managing Drupal projects.

Graphical development in a CMS like Drupal 7 is wonderful. It is possible to develop complex sites with very little coding. It is also great for getting started, especially for small sites. The anxiety begins when the site starts getting larger—and there's a set of developers, not just you alone.

As the complexity of the site increases, you need to separate the development environment from the production environment. You may want to have a separate testing environment as well. For development, you need minimal representative data. For testing, you may need simulated data in order to test for scalability and performance. The last thing you want, as a developer, is to be remembered for messing up live content—and the safest way to ensure that doesn't happen is to stay away from the operational site!

Since the data about the site and the content are in the same database, you may want to selectively back up and restore parts of the database. However, it is usually simpler to do the development, and when everything is working well, repeat the steps on the test and the deployment sites.

If the steps are being repeated, how do you ensure that no mouse click or field entry is different in the live site from the development site? And later, when multiple developers are involved, you will need to know who did what, and when.

How do you keep track of features being added to your site? It would be much simpler if the changes you make in Drupal 7 using the graphical interface could be saved as a text file, which could be used to implement the same changes in the test and deployment servers. You could further implement source code control using a repository like Git or Subversion. This is what you may do with UML modelling tools like *ArgoUML*, or form design tools like *Glade*.

Since this need is hardly unique, in all probability, somebody must have implemented a solution—and, indeed, they have. The *features* module (<http://drupal.org/project/features>) is one such solution.

The features module

It is very easy to overlook this module. The name is very inappropriate for a Google search! Plus, it seems to add nothing to the development. The usage statistics are, however, impressive. Over 75,000 sites have installed it, and the number continues to grow with Drupal 7. You will find the current

statistics at <http://drupal.org/project/usage>.

I agree with the following comment in <http://codekarate.com/blog/top-ten-best-drupal-7-contrib-modules>: “If you aim to become a serious Drupal developer, you can't live without the *features* module. You may think of a feature as a use case. Or you may want to group all changes you made to a Drupal site that implement some user need.”

But what does that mean?

Your new requirement may be to add new vocabulary in the taxonomy module. The next step would be that you may require additional content types, which may use the additional vocabulary you have defined. Finally, this content needs to be viewed—for which you may need to define custom views. Subsequently, you will need to define menu links, permissions, etc.

After you have added the needed functionality, you are ready to define a new feature. Give it a suitable name, and edit components to add to this feature. It is remarkably simple. You will need to add the *strongarm* module in order to export the vocabulary.

When all the components have been added to the feature, export it as a *tar* file, which can then be deployed on any other system. You may want to go through a tutorial to understand the steps—for example, <http://www.ostraining.com/blog/drupal/features/>.

If you make changes to any of the components, or add components to your features, you can re-export the revised feature, and deploy the updated version on the other systems. The content of the *tar* file can be added to a repository like Git, and you have version control for the feature. If every capability of your site is part of some feature, your entire site is managed by version control. The code for the site is now in the repository and distinct from the content.

You can choose to apply (or ignore) the advice and methodologies of software engineering while developing Drupal 7 projects as well—but you won't be able to say that it can't be done! 

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